Further developments of generalized Jeffreys's approximate Bayes factor

One-sided eJAB, regularity and Equally Weighted Decision Error Profiles

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Abstract

We further develop and exposit generalized Jeffreys's approximate Bayes factor (eJAB; Velidi and Wei, 2025). Equally Weighted Decision Error Profiles compare eJAB with test-statistic Bayes factors (TSBF; Johnson, 2005; Johnson et al., 2023), BIC, and NHST significance rules. For 2×2 and 3×3 Pearson chi-squared independence tests, eJAB generally outperforms estimated-ω TSBF outside small-sample or weak-signal regimes; the trend reverses for 5×5 tables. With true ω, eJAB generally matches or underperforms TSBF. BIC differences are small for single-parameter tests; eJAB gains at weaker effects in multiparameter tests, whereas BIC benefits at stronger effects. Against $p < .05$, eJAB gains with larger samples and stronger signals; weak signals and some small-sample settings favor NHST. We derive a null-centered one-sided eJAB via the Marsman–Wagenmakers correction and weaken regularity conditions while retaining consistency. Three clinical examples illustrate significant findings favoring the null.

One-sided eJAB

For a scalar parameter, compare $\mathcal{H}_0 : \theta = \theta_0$ with $\mathcal{H}_+ : \theta > \theta_0$.

Null-centered encompassing prior

Let $v_n = I_n^{-1}(\bar{\theta})$. Use a symmetric prior centered at θ_0 and the Gaussian posterior approximation:

$$\theta \mid \mathcal{H}_1 \sim \mathcal{N}(\theta_0, n v_n), \qquad \theta \mid \boldsymbol{y}, \mathcal{H}_1 \dot{\sim} \mathcal{N}(\bar{\theta}, v_n).$$

With compatible nuisance priors, the Savage–Dickey ratio gives

$$BF_{01}^{(0)} = \frac{\phi(\theta_0; \bar{\theta}, v_n)}{\phi(\theta_0; \theta_0, n v_n)} = \sqrt{n}\, e^{-W/2}, \qquad W = \frac{(\bar{\theta} - \theta_0)^2}{v_n}.$$

Directional mass correction

The prior mass in the predicted direction is $1/2$. For a moderate sample and relatively vague prior,

$$\Pr(\theta > \theta_0 \mid \boldsymbol{y}, \mathcal{H}_1) \approx 1 - p_1, \qquad BF_{+1} \approx 2(1 - p_1).$$

Reconstruct $p = 2\min\{p_1, 1 - p_1\}$. Substituting $W = Q_{\chi^2_1}(1 - p)$ gives

$$eJAB_{0+} = \frac{\sqrt{n}\exp\{-\frac{1}{2}Q_{\chi^2_1}(1-p)\}}{2(1-p_1)}.$$

Finite-sample convention

The exponent omits $(n-1)/n$; the directional factor $1/\{2(1-p_1)\}$ is retained. This is the leading Gaussian approximation under the null-centered construction.

The unrestricted definition retains $1 - n^{-1/q}$ in its exponent. Its manuscript derivation uses an MLE-centered prior. The two constructions are distinguished explicitly.

Regularity and consistency

For fixed integer dimension $q \geq 1$, the unrestricted definition is

$$eJAB_{01} = \sqrt{n}\exp\left\{-\frac{1}{2}\left(1 - n^{-1/q}\right) Q_{\chi^2_q}(1-p)\right\}.$$

R1′. Left-tail calibration under $\mathcal{H}_0$:

$$\Pr(p_n \leq \delta \mid \mathcal{H}_0) \longrightarrow C\delta, \quad 0 < \delta \leq U_p.$$

Here $0 < U_p \leq 1$ and $0 < C \leq 1/U_p$ are constants.

R2 (weakened). Under a fixed alternative, $p_n \xrightarrow{P} 0$ and

$$D_{q,n} = \sqrt{n}\begin{cases} -p_n \log p_n, & q = 1, \\ p_n, & q \geq 2 \end{cases} \xrightarrow{P} 0.$$

Then

$$eJAB_{01} \xrightarrow{P} \infty \quad (\mathcal{H}_0), \qquad eJAB_{01} \xrightarrow{P} 0 \quad (\mathcal{H}_1).$$

Proof sketch

Under R1′, $W_n = Q_{\chi^2_q}(1 - p_n) = O_P(1)$. Thus $\log eJAB_{01} = \frac{1}{2}\log n - O_P(1) \xrightarrow{P} \infty$.

For $q \geq 2$, $Q_{\chi^2_q}(1-p) \geq Q_{\chi^2_2}(1-p) = -2\log p$. Together with the original small-p bound for $q = 1$, this gives

$$0 < eJAB_{01} \leq n^{1/(2n^{1/q})} D_{q,n}^{1-n^{-1/q}} \xrightarrow{P} 0.$$

These weaker conditions retain consistency under fixed alternatives. Local sequences $\theta_n = \theta_0 + c/\sqrt{n}$ may fail R2.

Equally Weighted Decision Error Profiles

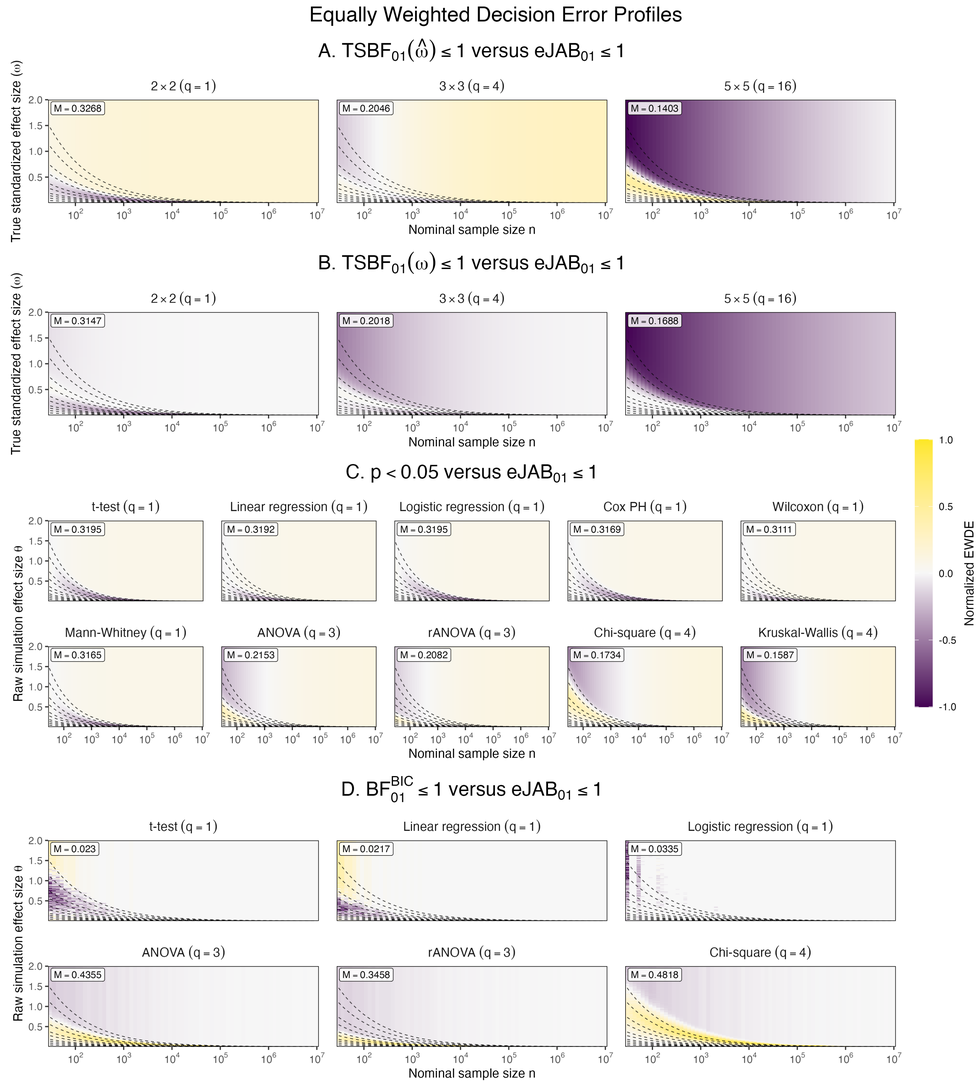
For a specified alternative, equally weighted decision error (EWDE) and its normalized comparison are

$$\mathrm{EWDE}_d(\theta, n) = \frac{\alpha_d + \beta_d}{2}, \qquad \mathrm{EWDE}_{\mathrm{norm}}(\theta, n) = \frac{\mathrm{EWDE}_{\mathrm{comparison}} - \mathrm{EWDE}_{eJAB}}{\max_{(\theta,n)\in\mathcal{G}} |\,\mathrm{EWDE}_{\mathrm{comparison}} - \mathrm{EWDE}_{eJAB}\,|}.$$

TSBF threshold and optimality. For fixed $\omega > 0$ in the chi-squared model, TSBF gives an upper-tail test that is uniformly most powerful among tests based on X^2 at its attained size (Neyman–Pearson). Cutoff 1 minimizes prior-predictive error under equal model probabilities and 0–1 loss. The same cutoff is used for eJAB; Panel A is a plug-in sensitivity analysis.

Panel A: plug-in effect estimate $\hat{\omega} = \sqrt{\frac{X^2}{n}}, \qquad \hat{\tau}^2 = \frac{n}{q}\hat{\omega}^2 = \frac{X^2}{q}.$

X^2: observed chi-squared statistic; n: sample size; q: degrees of freedom.



Equally Weighted Decision Error Profiles. A–B compare estimated- and true-effect TSBF with eJAB across three table sizes; C compares $p < .05$ across ten tests; D compares BIC across six parametric tests. Colors show normalized EWDE: yellow favors eJAB; purple favors the comparison. Each panel reports normalization M. Dashed $\theta = c/\sqrt{n}$ curves ($\theta = \omega$ in A–B) mark constant standardized signal; normalized EWDE can vary along them.

Clinical-trial contradictions

Three illustrative examples

Early Phase I

NCT00494806

First flatus: rocking vs standard care. t-test.

$n = 66, \quad p < .05$

$eJAB = 1.23$

Phase I

NCT03907683

Activity after a weekday move-more message. t-test.

$n = 80, \quad p = .037$

$eJAB = 1.04$

Phase II

NCT00539539

CPR fraction: feedback on vs off. Linear regression.

$n = 1{,}586, \quad p = .016$

$eJAB = 2.19$

For $p < .05$, eJAB uses the bound .05. Contradictions warrant follow-up; they do not establish Type I errors.

Paper on arXiv

Citations

Selected references Velidi and Wei (2025), *Generalized Jeffreys's approximate objective Bayes factor*, arXiv:2510.10358; Johnson (2005), *Bayes factors based on test statistics*, JRSS B 67, 689–701; Wagenmakers (2022), *Approximate objective Bayes factors from p-values and sample size*; Marsman & Wagenmakers (2017), *Educational and Psychological Measurement* 77, 529–539. Mulder, Wagenmakers & Marsman (2022), *The American Statistician* 76, 102–109; Johnson, Pramanik & Shudde (2023), *PNAS* 120, e2217331120; Fowlie (2021), *Neyman–Pearson lemma for Bayes factors.*